

DSCI 351 Recommender Systems

Assignment 3: Content-based recommender system

Dataset for assignment 3: `tmdb_5000_movies`, `tmdb_5000_credits`

What to submit:

- A pdf file includes only the answers to questions in each step.
- A Python code file.

Please follow the steps, and answer questions in each step.

Step 1. Load two files, observe the data, and merge them based on the common column(s). Perform data cleaning on `genres` feature first. How many unique genres are there in the dataset? List top 3 genres by count, provide the genre type and count value.

Step 2. Now look into the `cast` feature. Clean the column and find out how many unique casts in the dataset. List top 3 casts by count, provide the cast's name and their count value.

Step 3. Clean `keywords` feature, answer how many unique keywords in total? List top 3 key words and their count.

Step 4. Create a `director` column for each movie, by looking into `crew` feature. Who is the top ranked director by count?

Step 5. Vectorize the features: `genres`, `cast`, `director` and `keywords`. Use those four vectorized features, calculate the distance between any two movies. The distance between any two movies is the sum of those vectorized features' cosine distances: `genres_vector`, `cast_vector`, `director_vector` and `keywords_vector`. What is the distance between `movie_id = 285` and `movie_id=49026`? (round your answer to 3 decimal points)

Hint 1: Use [`scipy.spatial.distance.cosine`](#) to calculate the distance.

Hint 2: Vectorize the features: for example, suppose a full list of genres is `[Action, Adventure, Fantasy, Science Fiction, Crime, Drama]`, if a movie's genres are `[Action, Adventure, Crime]`, then the vector of this movie genres is `[1, 1, 0, 0, 1, 0]`.

Step 6. For any given movie id, found its top k neighbors. The larger the distance between the two movies, the less similar they are. For `movie_id=285`, provide its top 8 neighbors, with corresponding movie id and distance value.

Hint: do not include the movies in the neighborhood if distance is 0.

Step 7. Now make prediction for a movie's rating from its top k neighbors. For a given movie title:

- a. identify the corresponding movie id based on movie title
- b. get top k neighbor movies of the given movie (k=10)
- c. calculate the average `vote_average` of top k movies (k=10), use this mean value as the predicted rating of the given movie.

What is the predicted rating for movie: "Pirates of the Caribbean: At World's End", and compare it to the actual rating (`vote_average`) of the movie. Provide your observation and discussion.

Hint: If a movie's `vote_average` is 0, do not include it in the average calculation.

Note: This is not strict content-based recommendation for a specific user, as the `vote_average` is obtained from all users.